GODSWILL IMOLELE ANTHONY

Explainable and Uncertainty-Aware AI for Antimicrobial Discovery and Resistance
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Language Proficiency: English

EDUCATIONAL QUALIFICATIONS & STANDARDISED TESTING

Professional Diploma in Bioinformatics | Helix Biogen Institute, Nigeria — Apr 2022 – Jun 2023
Thesis: In silico comparative analysis of BRCA1 genes in selected mammals

Bachelor of Science (Biochemistry) | Ambrose Alli University, Ekpoma, Nigeria — 2014 – 2019
(WES-evaluated as a U.S. bachelor's degree equivalent) GPA: 2.9/4.0
Thesis: Effects of fermented African oil bean on plasma cholesterol in Wistar rats

GRE General Test: 331 — Quantitative 166 · Verbal 165 — Oct 2024

AWARDS, LEADERSHIP & SERVICE

- Global AMR R&D Hub Cross-Domain Award — 2026 Vivli AMR Surveillance Data Challenge (Argus AI; team lead). Sponsored by Pfizer, Johnson & Johnson, Paratek and the NIH GREI Innovation Award.
- Winner — 2025 Advanced CADD Machine-Learning Drug Discovery Challenge (leading enrichment-factor and error performance in an international West African cohort).
- Finalist, top-10% team — 2025 Vivli AMR Surveillance Global Data Challenge (ResistAI+).
- Oral Presentation Award — 2025 NextGen Africa Biomedical Science & Innovation Symposium (peptide-vaccine immunoinformatics).
- Team Lead, African Community for Systematic Review and Meta-Analysis — research-skills workshops for approximately 81 participants.

PROFESSIONAL AFFILIATIONS AND MEMBERSHIPS

- International Society for Computational Biology (ISCB) — 2024 to present
- Systematic Reviews Network (SRN), Nigeria — 2023 to present
- Bioinformatics Society of Nigeria — 2022 to present
- Nigeria Society for Biochemistry and Molecular Biology — 2022 to present

EMPLOYMENT HISTORY

Project Lead, Research and Teaching Assistant — Feb 2020 – Present
Helix Biogen Institute, Nigeria

- Built multi-omics cancer-survival models integrating transcriptomic, whole-genome, immune/HLA class I and clinical data; showed with Cox proportional-hazards modelling that low HLA class I expression predicts poorer clinical outcomes.
- Analysed RNA-seq and microarray data from Sri Lankan nasopharyngeal carcinoma patients (DESeq2, limma, GSEA) to identify prognostic genetic markers and immune-driven tumour signatures.
- Led a 4-month pancreatic-cancer metagenomics project with 9 undergraduate students, resulting in two manuscripts on the role of microbial genes in pancreatic cancer promotion and progression; identified key genes and explored their potential as therapeutic targets or biomarkers. Mentor to trainees in reproducible bioinformatics.
- Identified critical research gaps in Ebola Zaire strain diagnosis and led a collaborative project that selected and analysed immunogenic genes; designed a potential construct for a lateral flow diagnostic test kit.
- Led development of three AMR intelligence systems (Argus AI, ResistAI+ and NaijaResist; see Other Research Experiences).

Laboratory Research Assistant — Feb 2018 – Jul 2019
Dept. of Biochemistry, Nigerian Institute of Medical Research (NIMR), Nigeria

- Performed more than 200 ELISA and TIA assays and over 20 Western blot analyses, contributing to data reliability; supported nucleic-acid extraction and PCR.

- Managed and maintained multiple cohorts of mouse colonies, ensuring strict adherence to regulatory standards and Good Laboratory Practice (GLP), thereby maintaining the integrity of research outcomes.
- Streamlined experimental processes, leading to improved accuracy and consistency in assay results, contributing to more reliable data interpretation and decision-making.
- Contributed to research projects through careful experimental execution, supporting scientific publications and presentations.

OTHER RESEARCH EXPERIENCES

ML-Driven QSAR & Structural Bioinformatics for InhA Inhibitors | *First & corresponding author* 2026
Biology Methods and Protocols, 2026 — doi.org/10.1093/biomethods/bpag015

- Developed a machine-learning-driven QSAR and structural-bioinformatics workflow for natural-product screening against InhA, a tuberculosis target, integrating molecular descriptors, model evaluation, docking and molecular dynamics.

Argus AI — AMR Threat–Innovation Gap Observatory | *Team Lead, Helix Biogen Institute* 2026
Global AMR R&D Hub Cross-Domain Award — 2026 Vivli AMR Surveillance Data Challenge

- Bayesian hierarchical evidence synthesis of 4.95 million isolate–drug observations (71 countries, 44 species, ten antibiotics, 2004–2024) with global AMR R&D investment data; partial pooling plus isolate-weighted and country-balanced sensitivity analyses; traceable, source-grounded platform with versioned data lineage.

ResistAI+ — Bayesian AMR Surveillance & Forecasting | *Team Lead, Data Analysis & Model Deployment, Helix Biogen Institute* 2025
Finalist, top 10% of 100+ global teams — 2025 Vivli AMR Surveillance Global Data Challenge

- Led Bayesian hierarchical change-point modelling, MIC-creep analysis and posterior-predictive forecasting across heterogeneous multi-country data, with quantified uncertainty; deployed as an evidence-aware dashboard.

NaijaResist — Prototype AMR Intelligence System for Nigeria | *Project Lead & Architect, Helix Biogen Institute* 2026 – Present

- Machine-learning resistance prediction combined with genotype–phenotype reconciliation across phenotypic resistance, gene burden, cross-species transfer and organism-level threat, making discordance visible and reviewable.

MORPH-HSV — Resistance-Aware Prioritisation of HSV-1 DNA Polymerase Inhibitors | *Team Lead* 2026
2026 Shanghai Computational Biology Competition (preliminary round submitted)

- Led a four-person team ranking 100 non-nucleoside candidates for experimental testing against HSV-1 DNA polymerase, integrating evidence-guided active-space mining (~22,000 compounds), scaffold-aware machine learning, screening of ~5 million molecules, and Glide HTVS/XP re-docking against 8VQ2.
- Validated QSAR models on scaffold-separated splits (mean $R^2 = 0.55 \pm 0.07$ across ten randomised splits; 1,004 molecules, 541 scaffolds); final ranking combined model score, docking, synthetic accessibility, physicochemical and structural-alert gates, scaffold diversity, and re-docking against W781V, N815S and Y941H resistance variants where available.

BioSeq LTD — Bioinformatics Associate (Level 3) | *Remote, project-based* Oct 2026 – Present

- Support infection genomics, metagenomics and multi-omics projects: data curation, metadata harmonisation, statistical modelling, evidence synthesis and reproducible workflows.
- AGARI *E. coli* data integration (Africa CDC / ASLM): scientific contributor to BioSeq's proposal to integrate *E. coli* genomic surveillance into AGARI (African Genomes Archive for Response and Insights). Data-integration contributions:
  - Defined AMR results at the highest justified resolution (family, allele, mutation, mechanism) with call-resolution state and tool/database/version provenance; proposed PHA4GE hAMRonization mapping across AMR-detection tools.
  - Proposed genotype–phenotype discordance flags (both sources retained), method–drug–breakpoint compatibility checks, event-level lineage and synthetic acceptance cases.

RESEARCH OUTPUTS

1. **Anthony, G. I.**, et al. (2026). Discovery of novel InhA inhibitors through structural bioinformatics and machine learning-driven QSAR screening of natural products. *Biology Methods and Protocols*, 11(1), bpag015. doi.org/10.1093/biomethods/bpag015 [First and corresponding author]
2. Omoseeye, S. D., Azekhuoria, D. O., Owolade, A. J., Ibeh, S. J., **Anthony, G. I.**, Adaramola, I. M., Adeniyi, A. A., Iwuji, C. B., Adedara, J. F., & Bodun, D. S. (2026). Integrative machine learning and structural bioinformatics strategies for the identification of selective sterol 14α-demethylase inhibitors targeting *Trypanosoma cruzi*. *In Silico Research in Biomedicine*, 2, 100182. https://doi.org/10.1016/j.insi.2026.100182
3. **Anthony, G. I.**, Ogbozor, F. I., et al. (2025). In silico evaluation of 4-quinolone ligands interacting with GRK5 using message-passing neural networks, molecular docking and molecular dynamics. *Egyptian Journal of Medical Human Genetics*. doi.org/10.1186/s43042-025-00780-x [First author]
4. Lawal, O. P., Ahmed, N. K., Ilesanmi, T. A., **Anthony, G. I.**, et al. (2024). The impact of antimicrobial resistance on cancer treatment: current evidence and future directions. *Asian Journal of Research in Medical and Pharmaceutical Sciences*, 13(4), 9–27. doi.org/10.9734/ajrimps/2024/v13i4270
5. Ibiyemi, F. O., Awoniran, D., **Anthony, G. I.**, et al. (2025). Evidence-based management of urinary tract infections: balancing efficacy, safety and antimicrobial stewardship. *International Journal of Research in Pharmacy and Allied Science*, 4(10). doi.org/10.71431/IJRPAS.2025.41003
6. Abass, O. A., Timofeev, V. I., Sarkar, B., et al., including **Anthony, G. I.** (listed as "Imolele, A. G.") (2022). Immunoinformatics analysis to design a novel epitope-based vaccine candidate targeting the glycoprotein and nucleoprotein of Lassa mammarenavirus using strains from Nigeria. *Journal of Biomolecular Structure and Dynamics*, 40(16), 7283–7302. doi.org/10.1080/07391102.2021.1896387
7. Ibiyemi, F. O., **Anthony, G. I.**, et al. (2025). Antipsychotic medications and patient safety: a systematic analysis of adverse drug reactions across drug classes. *International Journal of Research in Pharmacy and Allied Science*, 4(10), 66–85.
8. Nwosu, S. N., **Anthony, G. I.**, et al. (2024). Tannin extraction and analysis in *Telfairia occidentalis* seeds: a TLC and UV-based approach. *Asian Journal of Medical Principles and Clinical Practice*, 7(2), 400–407. https://journalajmpcp.com/index.php/AJMPCP/article/view/248
9. Andowase, P., & **Anthony, G. I.** (2020). The effect of combined treatment of vitamin C and loperamide on intestinal sodium and potassium ion ATPase, alkaline phosphatase and lipid peroxidation on castor oil induced diarrheal rats. *bioRxiv* [preprint]. doi.org/10.1101/2020.11.13.380469

Indexed name variants (Imolele, A. G.; Anthony Goodwill Imolele) refer to the same author, ORCID 0000-0002-3093-3309.

ADDITIONAL INFORMATION/TECHNICAL SKILLS

- **AI for molecular discovery:** QSAR, message-passing neural networks, molecular descriptors, scaffold-aware model evaluation, molecular docking and dynamics (AutoDock Vina, Glide, PyMOL), structure modelling (GalaxyRefine, I-TASSER, Boltz-2).
- **Statistical & probabilistic modelling:** Bayesian hierarchical models, change-point analysis, posterior-predictive forecasting, sensitivity analysis, Cox proportional-hazards, uncertainty quantification.
- **Omics & data integration:** RNA-seq/microarray (DESeq2, limma, GSEA), multi-omics integration, NGS processing, metagenomics; metadata harmonisation, lineage and provenance.
- **Programming & deployment:** Python, R, SQL, Bash; Streamlit dashboards; reproducible pipelines and documentation.